ANNA UNIVERSITY, CHENNAI

B.E./B.Tech Degree Examination, April/May 2025

Computer Science and Engineering

CS3491 - Artificial Intelligence

Regulation 2025

Time : Three Hours | Maximum Marks : 100

Reg. No.

Answer ALL Questions

PART A (10 × 2 = 20)

1. Distinguish between utility-based agents and goal-based agents with a suitable example for each.
2. Verify if Breadth-First Search (BFS) is optimal when the step costs are not identical but are positive. Justify your answer.
3. Define the concept of Arc Consistency (AC-3) in Constraint Satisfaction Problems.
4. Why does Alpha-Beta pruning yield the exact same optimal decision as minimax search despite bypassing several branch
5. Translate the following assertion into First-Order Logic: Every student who takes the Artificial Intelligence course loves pro
6. Define unification and explain its significance in First-Order Logic inference.
7. Contrast progression planning with regression planning in classical planning systems.
8. State the conditional independence relationships implied by a Bayesian Network topology using the local Markov conditio
9. Define inductive bias and explain why a learning algorithm must possess it.
10. In reinforcement learning, explain why an agent cannot rely solely on exploitation to maximize its long-term reward.

PART B (5 × 13 = 65)

11. (a) Demonstrate the operation of the A* search algorithm. Explain how the properties of heuristic admissibility and heuristic

OR

(b) (i) Formulate the classic 8-puzzle game as a state-space search problem, clearly detailing the states, initial state, actions, t

12. (a) Consider a game tree where the root is a MAX node, branching down to three children (A, B, C). A has leaf descendan

OR

(b) Discuss the backtracking search algorithm for solving Constraint Satisfaction Problems (CSPs). Explain how the search eff

13. (a) (i) Convert the following sentences into Conjunctive Normal Form (CNF):□1. Anyone who loves animals is loved by son

OR

(b) Describe the architecture and reasoning mechanism of a Knowledge-Based Agent. Explain and compare the forward chain

14. (a) Discuss Classical Planning using the Planning Domain Definition Language (PDDL). Write a complete PDDL represent

OR

(b) Define a Bayesian Belief Network. Explain the exact inference algorithm using variable elimination. Demonstrate its applica

15. (a) Explain the concept of decision tree induction using the ID3 algorithm. Show how information gain and entropy are mat

OR

(b) (i) Explain the architectural design, activation function, and learning rule of a single-layer Perceptron. Detail its limitation reg